Topology Control in 3-Dimensional Networks & Algorithms for Multi-Channel Aggregated Convergecast

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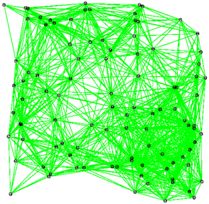
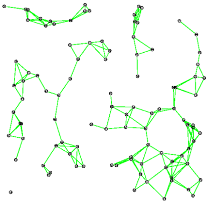
Why Topology Control?

Goal: Reduce Power Consumption

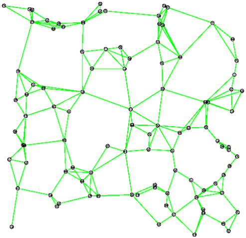
Given a network connectivity graph, compute a subgraph with certain properties: connectivity, spanner, low interference, etc.





Leads to: High energy consumption, High interference, Low throughput, Partitioned network

Why Topology Control?



Benefits

1. Global connectivity
2. Low energy consumption
3. Low interference
4. High throughput

Problem

To find optimal transmission power levels using local information such that network connectivity is maintained.

3-Dimensional Wireless Networks

Challenges

1. Node density if prohibitively high to guarantee connectivity
   - Critical avg. node degree (d): 15 in 2D, 34 in 3D for $n = 1000$ [Poduri, EmNets '06]
2. Many 2D algorithms not readily extensible to 3D (e.g., geographic routing)
3. No ordering of nodes based on angular information
4. High complexity

Applications

Structural health monitoring, Marine-life monitoring (Underwater networks)

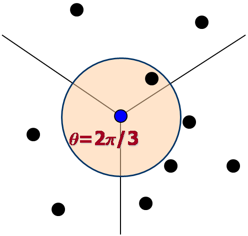
Cone Based Topology Control

2D CBTC

1. Global connectivity from local geometric constraints
   [Wattenhofer, Infocom '01], [Li Li, PODC '01]
2. Maximum Power Graph $G = (V, E)$ is connected
3. Receivers can determine direction of senders, needs node ordering
4. Complexity $O(d \log d)$, $d =$ average node degree

Main Result: The θ constraint

If every node adjusts its power level so that there exists at least one neighbor at every $\theta = 2\pi/3$ sector around itself, then the network is connected.



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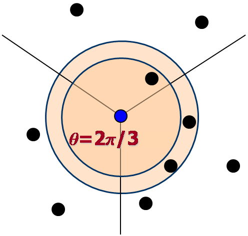

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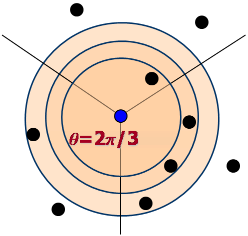
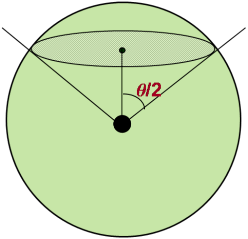


Cone Based Topology Control

3D CBTC

1. Each node increases its power level until there is at least one neighbor at every 3D cone of apex angle $\theta = 2\pi/3$ around it
   [Bahramgiri, ICCCN '05, Wireless Networks '06]
2. Assumes directional information
3. High time complexity $O(d^3 \log d)$



Approach

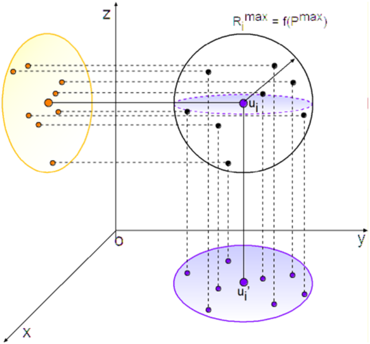
2 Phases

1. Phase 1
   - Use Multi-Dimensional Scaling (MDS) to find relative location maps for each nodes neighbors when they use P^{max}
     Input: Pairwise distances
     Output: Relative locations
2. Phase 2
   - Using orthographic projections (convert the 3D problem into similar problems in 2D)
     hspace3cmOR
   - Using Spherical Delaunay Triangulation (SDT)

Orthographic Projection

Algorithm

1. Each node starts with minimum tx power
2. For a given tx power, project the neighbors on *xy*, *yz*, and *zx* plane
   - If any of the 3 planes does not satisfy the $\theta = 2\pi/3$ constraint, increase power to the next level
   - Else STOP, settle with current power
3. Run 2D CBTC on each plane
4. Go back to Step 2 unless P^{max} is reached



Intuition

Satisfying 2D CBTC on 3 planes $\Rightarrow$ non-empty 3D cones with apex angle $2\pi/3$

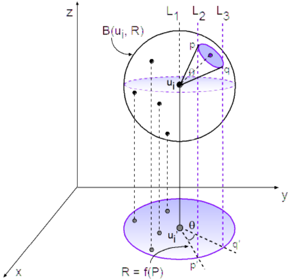
Orthographic Projection

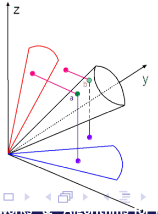
Lemma

Consider the projections of a node u_i's neighbor on the three planes xy, yz, and zx.
If there exists at least one empty sector of angle θ in any one of the planes, then there exists an infinite number of empty 3D cones with apex angle θ around u_i.

Connectivity?

Theoretically NO, practically YES in almost all cases

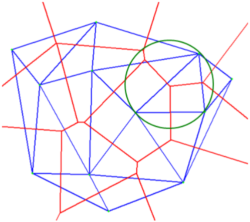


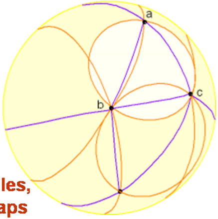


Spherical Delaunay Triangulation

Delaunay Triangulation

1. Dual of a Voronoi diagram
2. Given N points in 2D, a Voronoi diagram tessellates the plane into N convex polygons such that any point in a polygon is closest to the point that lies in that polygon
3. Empty circum-circle property

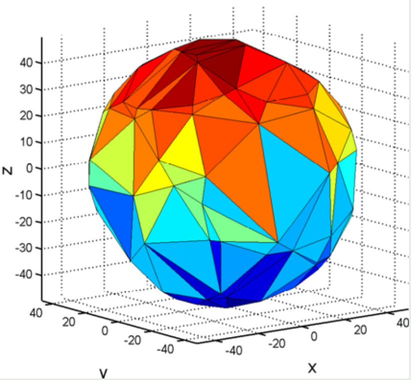




Spherical triangles, and spherical caps

Spherical Delaunay Triangulation

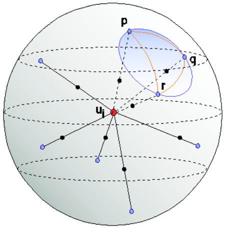
SDT using Quickhull for 100 points randomly distributed on the surface of a sphere of radius 50



Spherical Delaunay Triangulation

Algorithm SDT

1. Each node starts with minimum tx power
2. For a given tx power, project the neighbors on the spherical surface
3. Construct Delaunay triangulation on the surface of the sphere
4. Calculate the area of the (empty) spherical caps
5. If any cap area is $> 2.7R^2$, increase the power to next level; go to Step 2
6. Else, stop and settle down with current power

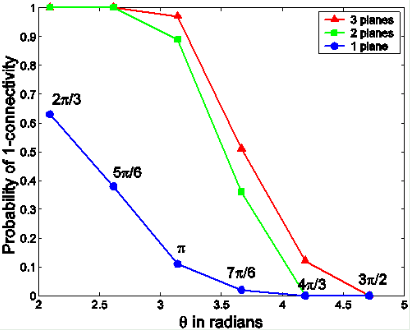


Lemma

If none of the spherical caps have a surface area greater than $2.7R^2$, the network is at least one-connected

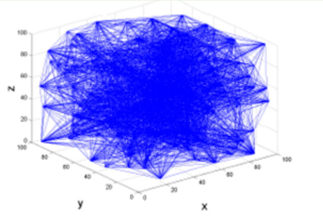
Orthographic Projection

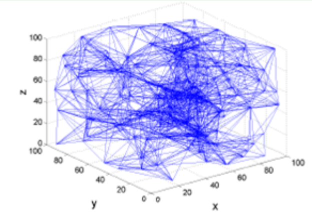
Probability of network connectivity as the θ constraint is satisfied on 1, 2, or all 3 planes for 100 nodes

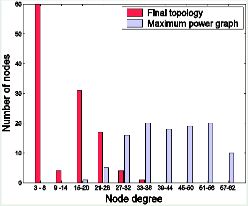


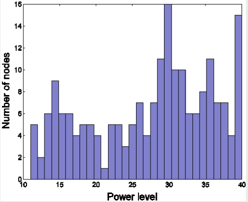
SDT

Node deg of max power graph vs. final topology: $n = 200$, $P^{max} = 40$



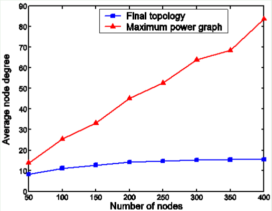


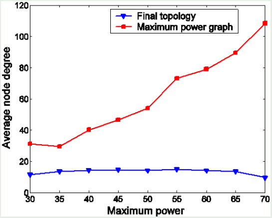




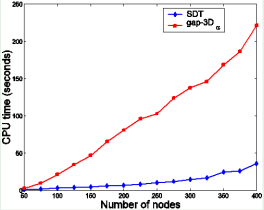
SDT

Avg node deg vs. network size and max power





Comparison of CPU execution times with $P^{max} = 40$



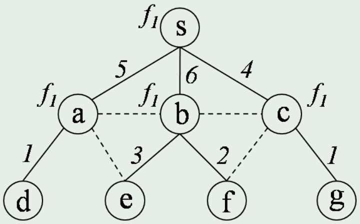
Part II

Algorithms for Multi-Channel Fast Aggregated Convergecast in Sensor Networks

Joint work: Amitabha Ghosh, Ozlem Durmaz Incel, V. S. Anil Kumar, Bhaskar Krishnamachari

An Illustration

Single Frequency

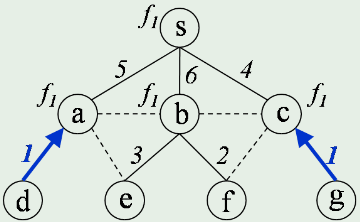


Setting

1. Half-duplex transceiver: cannot transmit and receive simultaneously
2. Single transceiver: cannot receive multiple packets simultaneously
3. Interference causes packet loss
4. Receiver-based frequency assignment

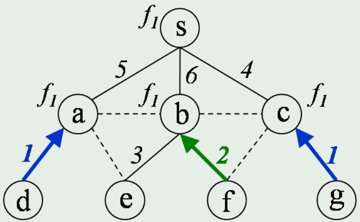
An Illustration

Single Frequency



An Illustration

Single Frequency



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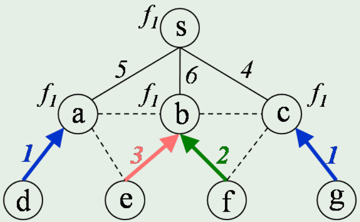
Single Frequency



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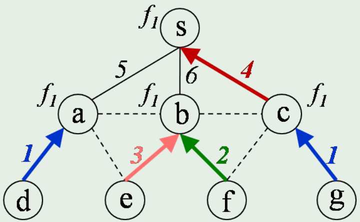

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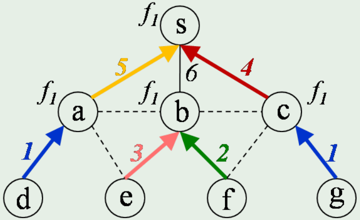

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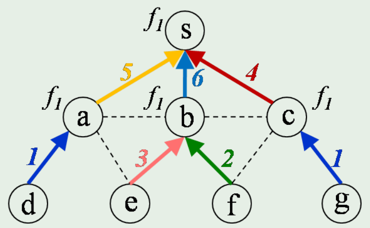

An Illustration

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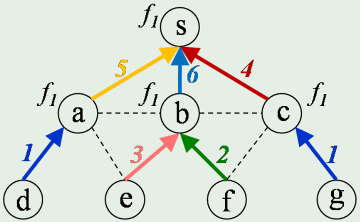
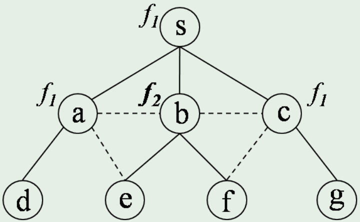


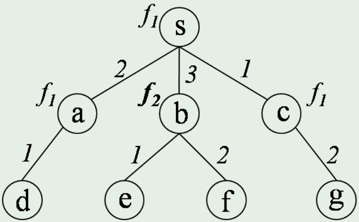
Multiple Frequencies



	Frame 1 (Aggregated data)			Frame 2 (Aggregated data)		
Receiver	Slot 1	Slot 2	Slot 3	Slot 1	Slot 2	Slot 3
s						
a						
b						
c						

An Illustration

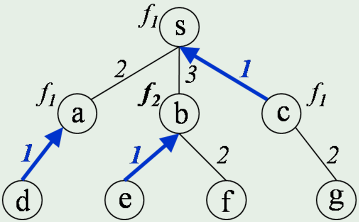
Multiple Frequencies



	Frame 1 (Aggregated data)			Frame 2 (Aggregated data)		
Receiver	Slot 1	Slot 2	Slot 3	Slot 1	Slot 2	Slot 3
s	c					
a	d					
b	e					
c						

An Illustration

Multiple Frequencies



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s	c	a, d				
a	d					
b	e	f				
c		g				

An Illustration

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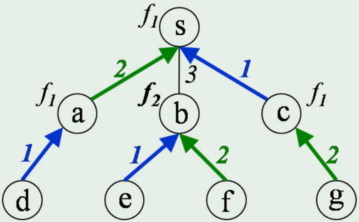

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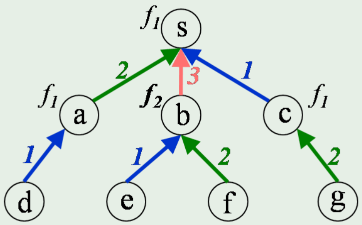

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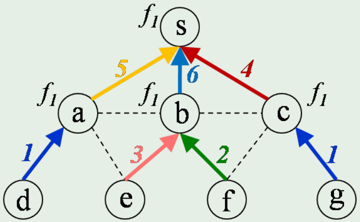
s	c	a, d	b, e, f	c, g	a, d	b, e, f
a	d			d		
b	e	f		e	f	
c		g			g	

An Illustration

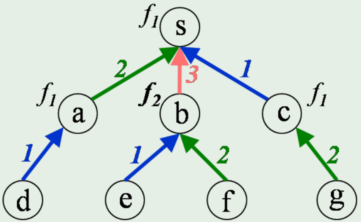
1 Frequency, 6 slots



Question

What is the fastest rate at which aggregated data can be collected from the network? (minimizing the schedule length)

2 Frequencies, 3 slots



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Receiver	Slot 1	Slot 2	Slot 3	Slot 1	Slot 2	Slot 3
s	c	a, d	b, e, f	c, g	a, d	b, e, f
a	d			d		
b	e	f		e	f	
c		g			g	

Optimal Frequency Assignment

MFAP: Minimum Frequency Assignment Problem

Given a spanning tree T on an arbitrary graph $G = (V, E)$, find the minimum number of frequencies that could be assigned to the receivers of T such that all the interfering link constraints are removed.

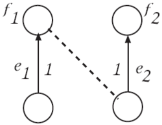


Figure: Interfering edge structure

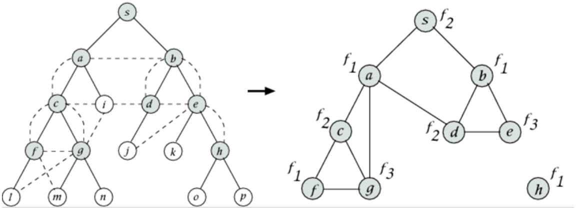
Theorem

MFAP is NP-complete.
Proof: Reduction from Vertex Color.

An Upper Bound

Lemma

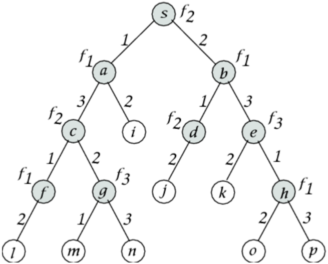
Construct a constraint graph $G_C = (V_C, E_C)$ *from* $G = (V, E)$ *as follows:*
For each receiver $v_i \in G$*, create a vertex* $u_i \in G_C$*. Create an edge between two such vertices* u_i *and* u_j *if their corresponding receivers are part of an interfering edge structure.*
Then, the number of frequencies that will remove all the interfering link constraints is: $K_{max} \leq \Delta(G_C) + 1$*, where* $\Delta(G_C)$*: maximum degree in* G_C*.*



BFS Time Slot Assignment

Theorem

Algorithm BFS-TimeSlotAssignment *on a tree gives the minimum schedule length equal to* $\Delta(T)$.



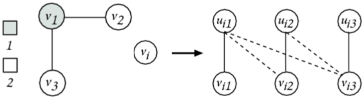
Minimizing the Schedule Length

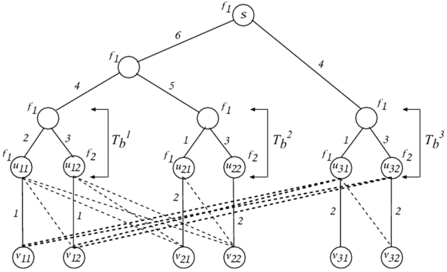
MFMTSP: Multiple-Frequency Minimum Time Scheduling Problem

Given a spanning tree T on an arbitrary graph $G = (V, E)$ and a constant number q of frequencies, find an assignment of frequencies and time slots such that the schedule length is minimized.

Theorem

MFMTSP is NP-complete.
Proof: Reduction from Vertex Color.



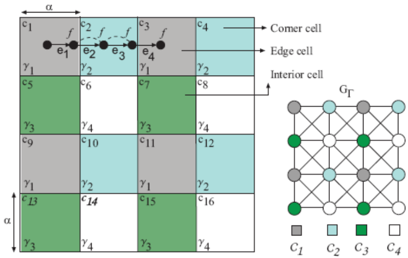


Time Slot Assignment on UDG

Lemma

Suppose γ_{c_i} denote the set of time slots to schedule the edges in cell c_i. Then, the minimum schedule length Γ for the whole network is:

$$\Gamma \leq 4 \cdot \max_{c_i} \{|\gamma_{c_i}|\}, \forall \alpha \geq 2.$$



Frequency Assignment on UDG

Load-Balanced Frequency Assignment

$R_{c_i} = \{v_1, \ldots, v_n\}$: set of receivers on T
$m : R_{c_i} \to \{f_1, \ldots, f_K\}$: mapping of frequencies to the receivers

Define load on f_k under m in c_i as:

$$l_{c_i}^m(f_k) = \sum_{v_j \in R_{c_i}, m(v_j)=f_k} deg^{in}(v_j)$$

Then, a load-balanced frequency assignment m^* in c_i is

$$m^* = \arg\min_m \max_{f_k} \{l_{c_i}^m(f_k)\}$$

Lemma

Load-balanced frequency assignment is NP-complete. [Garey, Johnson]

Frequency Assignment on UDG

Algorithm FrequencyGreedy

(1) Sort the nodes in R_{c_i} in non-increasing order of their in-degrees. Let this order be: $deg^{in}(v_1) \geq deg^{in}(v_2) \geq \ldots \geq deg^{in}(v_n)$.

(2) Starting from v_1, assign each successive node a frequency from $\{f_1, \ldots, f_K\}$ that has the least load on it so far, breaking ties arbitrarily.

Lemma

Algorithm FrequencyGreedy *in each cell c_i gives a $(4/3 - 1/3K)$ approximation on the optimal load $L_{c_i}^{m^*}$.*

Proof: Follows from Graham's algorithm for job scheduling on identical machines according to longest processing time first (LPT).

An Upper Bound on Time Slots per Cell

Lemma

If $L^{\phi}_{c_i}$ denote the load on the maximally loaded frequency in c_i under mapping $\phi : R_{c_i} \rightarrow \{f_1, \ldots, f_K\}$ achieved by FrequencyGreedy, *then any greedy time slot assignment can schedule all the edges in c_i within $2L^{\phi}_{c_i}$ time slots, i.e.,*

$$|\gamma_{c_i}| \leq 2L^{\phi}_{c_i}.$$

A Constant Factor Approximation

Theorem

Given a tree T on a UDG G, and K frequencies, there exists an algorithm $\mathcal{G}$ that achieves a constant factor $8\mu_\alpha\,(4/3 - 1/3K)$ approximation on the optimal schedule length, where $\mu_\alpha > 0$ is a constant for a given cell size $\alpha \geq 2$.

Proof Sketch: $\mathcal{G}$ consists of 2 phases:

(1) Run FREQUENCYGREEDY in each cell c_i
(2) Run any greedy time slot assignment scheme, e.g., greedily schedule a maximal number of edges at each slot
Lower bound on OPT: $\Gamma_{OPT} \geq \max_{c_i}\{L_{c_i}^{m^*}\}/\mu_\alpha$

$$\begin{aligned}
\Gamma_{\mathcal{G}} &\leq 4 \cdot \max_{c_i}\{|\gamma_{c_i}|\} \leq 8 \cdot \max_{c_i}\{L_{c_i}^{\phi}\} \\
&\leq 8 \cdot \max_{c_i}\{(4/3 - 1/3K) \cdot L_{c_i}^{m^*}\} \\
&\leq 8\mu_\alpha\,(4/3 - 1/3K) \cdot \Gamma_{OPT}
\end{aligned}$$

Assignment on Arbitrary Trees Under UDG

Theorem

Given a UDG G and K frequencies, there exists an algorithm $\mathcal{H}$ that achieves a constant factor $8\mu_\alpha\Delta_C$ approximation on the optimal schedule length, where $\mu_\alpha > 0$ is a constant for a given cell size $\alpha \geq 2$, and $\Delta_C > 0$ is a constant.

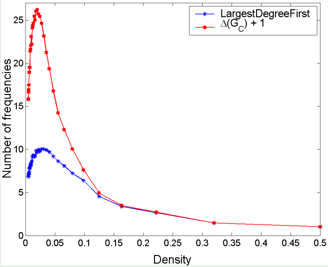
Proof Sketch:
V_{c_i}: set of nodes in c_i; R_{c_i}: set of receivers in c_i for an arbitrary T
Lower bound on OPT: $\Gamma_{OPT} \geq \max_{c_i}\{\lceil |V_{c_i}|/K \rceil\}/\mu_\alpha$
Run a CyclicFrequencyAssignment: $\psi(v_i) = i \mod K$ if $i \neq qK$, and K otherwise, $q \in \mathbb{N}^+$

Frequency Bounds

Largest Degree First (LDF)

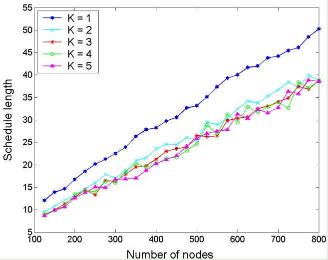
1. Input: Constraint Graph $G_C = (V_C, E_C)$
2. while $V_C \neq \phi$ do
3. $u \leftarrow$ vertex with maximum degree in V_C
4. Assign the first available frequency to u different from its neighbors
5. $V_C \leftarrow V_C \setminus \{u\}$

Comparison of LDF and $\Delta(G_C) + 1$



Schedule Length

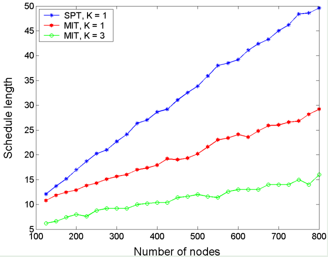
Alogrithm $\mathcal{G}$ on Shortest Path Tree for Different Network Sizes



Square region of size 200 $\times$ 200
Maximum transmission range 25

Schedule Length

Alogrithm $\mathcal{G}$ on Minimum Interference Tree for Different Network Sizes



Square region of size 200×200
Maximum transmission range 25